

EDUCATION

Purdue University <i>B.Sc. Computer Engineering Technology, Minor in Information Technology</i> ◦ GPA: 3.22 (Major: 3.53) Selected for Purdue’s competitive SoCET program: hands-on, industry-quality SoC design flow.	West Lafayette, IN 08/2023 - 05/2027
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PROJECTS

SoCET VLIW AI Accelerator “Atalla” - Compilers Team ◦ Spearheaded development of a CI pipeline and test scripts verifying every build stage from PyTorch kernels through graph compiler to AtallaC compiler to functional simulator, creating performance metrics to quantify code improvements on each compiler revision. ◦ Root-caused inefficient VLIW packetization to upstream register allocation optimized for serial execution by a greedy list scheduler, which induces artificial hazards for parallel operations and pinpoints the register allocator as the fix point. ◦ Drove cross-repository consolidation across 3 repositories and 9 branches, replacing stale code copies with versioned Git submodule dependencies so performance improvements flow into the build pipeline for compounding system-level gains.	08/2026 - Present
Senior Design Capstone - 4-Layer FPGA Expansion Shield ◦ Iterating on a 4-layer expansion shield PCB for Altera FPGA lab boards: ADC, DAC, audio codec, joystick, LEDs/switches, PMOD and SMA expansion, jumper configuration, and bench test points. ◦ Refining a prior team’s layout against customer requirements to improve fit, connector access, and bench debuggability.	08/2026 - Present
Modernized Superheterodyne FM Radio Receiver ◦ Engineered embedded C firmware for Arduino Mega digitally tuning a superheterodyne FM receiver via I2C 12-bit DAC, ADC RSSI feedback, and LCD interface with autonomous station acquisition across 88-108 MHz. ◦ Characterized VCO behavior using quartic regression over 56 measured data points ($R^2 = 0.9987$), enabling accurate digital frequency estimation and a DSP-assisted channel scan algorithm computing noise floor and identifying RSSI peaks.	04/2025

EXPERIENCE

Purdue University <i>Teaching Assistant - Embedded Digital Systems</i> ◦ Mentored 20 students in embedded C firmware: interrupts, timers, GPIO, and peripheral interfaces. ◦ Debugged student firmware and breadboard wiring with oscilloscopes, isolating circuit, timing, and logic faults. ◦ Recommended by the course professor (associate department head, ABET ETAC commissioner) and the graduate TA.	West Lafayette, IN 08/2026 - Present
Wistron <i>NVIDIA Server Repair Technician</i> ◦ Processed 3 NVIDIA HGX 8-GPU AI server baseboards per day end to end: intake, physical verification, test suites, FPGA telemetry decoding, re-test, and QC hand-off, isolating PCIe signal-integrity root causes. ◦ Engineered Python automation tooling that eliminated thousands of lines of manual database entry, saving 80+ labor hours weekly and streamlining the repair operations workflow in ETQ Reliance. ◦ Validated PCIe 5.0 retimers and NVSwitches via firmware flashing and interop test suites to verify system-level functionality.	Dallas, TX 05/2024 - 08/2024
Purdue Electric Racing <i>Embedded Systems Team Member</i> ◦ Architected a register-level PWM HAL for STM32 vehicle subsystems, cutting fan-control code from ~40 lines to 2. ◦ Refined vehicle CAN topology by isolating high-throughput motor traffic onto a dedicated bus to improve stability.	West Lafayette, IN 09/2024 - 12/2025
Jawa.gg <i>Verified Seller - PCs & Components</i> ◦ Reverse-engineered board-level failures using schematics and BoardView files; completed BGA rework and precision micro-soldering. ◦ Ranked Top 10 platform-wide in seller performance and recognized by CEO Amanda Stefan for contributions to marketplace growth.	Remote 10/2021 - 08/2023

SKILLS

Languages: C, Python, Java, VHDL, TypeScript, Structured Text Hardware/Chip Design: Digital Logic, FPGA (Quartus), PCB Design (KiCad), Mixed-Signal Circuits, SolidWorks Testing/Validation: CI/CD, Automated Testing, Functional Simulation, Lab Validation, Bring-Up, Root Cause Analysis Embedded/Firmware: Bare-Metal C, FreeRTOS, STM32, Arduino, Raspberry Pi Pico, HAL, Microcontroller Peripherals AI/ML & Compilers: AI Accelerator Toolchain, VLIW Optimization, Compiler Architecture, Hardware-Software Co-Design Systems/Modeling: Performance Benchmarking, Regression Modeling, MATLAB, Simulink, Multisim, Data Analysis Interfaces/Protocols: I2C, SPI, UART, BLE, MQTT, CAN, GPIO, PWM, ADC, DAC, USB Linux/Systems: Linux Administration, Bash Scripting, KVM, LXC, Proxmox, ZFS Development Tools: Git/GitHub, PlatformIO, STM32Cube, Linux, Selenium Standards: ISO 26262, IEC 61131-3 Lab Instrumentation: Oscilloscopes, Logic Analyzers, DAQ, Spectrum Analyzer, Power Supplies, BGA Rework Interpersonal: Problem Solving, Analytical Thinking, Cross-Functional Collaboration, Communication, Mentoring, Adaptability
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